

1. THE NATURE OF INSTRUCTIONAL MEDIA

The term media was first used to describe newspapers more than two centuries ago. Today media has many different connotations. For instance, there are mass media, print media, visual media and social media. While media can take on many different forms, the purpose of all media is universally the same -- media is a channel of communication.

Media are the means for transmitting or delivering messages and in teaching-learning perspective delivering content to the learners, to achieve effective instruction. Media can be used effectively in formal situation where students are working independently or teacher is working with other group of students.

The term instructional media has been defined in a variety of ways. In some cases, it refers to all aids that are used by the lecturer and students. In other cases, it refers only to printed media. In this chapter, we shall use the term to mean all devices and materials used in the teaching and learning processes. This definition is close to a broader definition such as that of Romiszowski (1981: 339) which includes not only electronic communications media, but also such devices as slides, photographs, teacher-made diagrams, charts, real objects and handouts that we use in the process of planned instruction. Figure 6.1 gives examples of some instructional media.

Type	Examples
Print	Pamphlets, handouts, study guides, manuals
Visual	Charts, real objects, photographs, transparencies
Audiovisual	Slides, tapes, films, filmstrips, television, video, multimedia
Static/display	Chalkboard, felt board, display easels, flip charts, cloth board, magnetic board
Electronic	Radio computers, electronic mail, CD-ROM, multimedia

2. LOOKING AT MEDIA

Media (plural of medium) is another word that has many definitions and I will argue that it has two distinct meanings relevant for teaching and learning, both of which are different from definitions of technology.

The word 'medium' comes from the Latin, meaning in the middle (a median) and also that which intermediates or interprets. Media require an active act of creation of content and/or communication, and someone who receives and understands the communication, as well as the technologies that carry the medium.

Media linked to senses and 'meaning'.

We use our senses, such as sound and sight, to interpret media. In this sense, we can consider text, graphics, audio and video as media 'channels', in that they intermediate ideas and images that convey meaning. Every interaction we have with media, in this sense, is an interpretation of reality, and again usually involves some form of human intervention, such as writing (for text), drawing

or design for graphics, talking, scripting or recording for audio and video. Note that there are two types of intervention in media: by the ‘creator’ who constructs information, and by the ‘receiver’, who must also interpret it.

Media of course depend on technology, but technology is only one element of media. Thus we can think of the Internet as merely a technological system, or as a medium that contains unique formats and symbol systems that help convey meaning and knowledge. These formats, symbol systems and unique characteristics (e.g. the 140-character limit in Twitter) are deliberately created and need to be interpreted by both creators and end users. Furthermore, at least with the Internet, people can be at the same time both creators and interpreters of knowledge.

Computing can also be considered a medium in this context. I use the term computing, not computers, since although computing uses computers, computing involves some kind of intervention, construction and interpretation. Computing as a medium would include animations, online social networking, using a search engine, or designing and using simulations. Thus Google uses a search engine as its primary technology, but I classify Google as a medium, since it needs content and content providers, and an end user who defines the parameters of the search, in addition to the technology of computer algorithms to assist the search. Thus the creation, communication and interpretation of meaning are added features that turn a technology into a medium.

Thus in terms of representing knowledge we can think of the following media for educational purposes:

- Text
- Graphics
- Audio
- Video
- Computing

Within each of these media, there are sub-systems, such as

- **text:** textbooks, novels, poems
- **graphics:** diagrams, photographs, drawings, posters, graffiti
- **audio:** sounds, speech
- **video:** television programs, YouTube clips, ‘talking heads’
- **computing:** animation, simulations, online discussion forums, virtual worlds.

Furthermore, within these sub-systems there are ways of influencing communication through the use of unique symbol systems, such as story lines and use of characters in novels, composition in photography, voice modulation to create effects in audio, cutting and editing in film and television, and the design of user interfaces or web pages in computing. The study of the relationship between these different symbol systems and the interpretation of meaning is a whole field of study in itself, called semiotics.

In education we could think of classroom teaching as a medium. Technology or tools are used (e.g. chalk and blackboards, or PowerPoint and a projector) but the key component is the intervention of the teacher and the interaction with the learners in real time and in a fixed time and place. We can also then think of online teaching as a different medium, with computers, the Internet (in the sense of the communication network) and a learning management system as core technologies, but it is the interaction between teachers, learners and online resources within the unique context of the Internet that are the essential component of online learning.

From an educational perspective, it is important to understand that media are not neutral or 'objective' in how they convey knowledge. They can be designed or used in such a way as to influence (for good or bad) the interpretation of meaning and hence our understanding. Some knowledge therefore of how media work is essential for teaching in a digital age. In particular, we need to know how best to design and apply media (rather than technology) to facilitate learning.

Over time, media have become more complex, with newer media (e.g. television) incorporating some of the components of earlier media (e.g. audio) as well as adding another medium (video). Digital media and the Internet increasingly are incorporating and integrating all previous media, such as text, audio, and video, and adding new media components, such as animation, simulation, and interactivity. When digital media incorporate many of these components they become 'rich media'. Thus one major advantage of the Internet is that it encompasses all the representational media of text, graphics, audio, video and computing.

Media as Organizations

The second meaning of media is broader and refers to the industries or significant areas of human activity that are organized around particular technologies, for instance film and movies, television, publishing, and the Internet. Within these different media are particular ways of representing, organizing and communicating knowledge.

Thus for instance within television there are different formats, such as news, documentaries, game shows, action programs, while in publishing there are novels, newspapers, comics, biographies, and so on. Sometimes the formats overlap but even then there are symbol systems within a medium that distinguish it from other media. For instance, in movies there are cuts, fades, close-ups, and other techniques that are markedly different from those in other media. All these features of media bring with them their own conventions and assist or change the way meaning is extracted or interpreted.

3.OPERATES OF MEDIA

Media can be used in direct instruction, active learning teaching strategies and student projects.

- Existing media resources can be used within **lectures** to stimulate interest in and develop knowledge of the material being taught. This traditional approach is teacher-centric, and information is pushed to the learner. Media allows the instructor to facilitate the transfer of expert knowledge to novice learners. Given the tremendous rate of technological change, instructors face an ongoing challenge in choosing the most effective media platform to reach

their students. Instructors can also create their own media to effectively and efficiently convey knowledge.

- Existing media resources can also be used to engage students and facilitate **active learning** strategies which promote deeper learning. For example, media provides a useful platform for teaching with cases, cooperative learning, problem solving, and for giving more interactive lecture demonstrations.
- **Student-created media** involves a high degree of engagement; promotes individual learning, social interaction and immersion; and is highly customizable and collaborative (Yowell and Rhoten, 2009). Student-created media provide an alternative or a complement to traditional undergraduate student research. By doing a digital storytelling project, personal reflection and communication by students can be promoted.

4. THE MANIPULATIVE PROPERTY

Manipulatives in the content of education are physical tools of teaching, engaging students visually and physically with objects, such as coins, blocks, puzzles, markers etc. The use of manipulatives is constructivist because students are actively engaged in discovery during the learning process.

Manipulatives can also be tools to help students solve problems. By using physical models to represent their thinking, they can move and adapt the materials as they explore possible solutions to problems. In real life, many people use models to help solve problems, such as an architect who might construct a model of a building or an engineer who might build a prototype of a piece of equipment.

For many students, concrete materials provide support in dealing with a subject that can be difficult and confusing. Students begin learning about a concept using manipulatives and progress to recording their work with manipulatives. Students connect their constructions to the written record of symbols and numbers. Concrete materials are imperative for exploration and experimentation with math ideas as students develop meaning. We want all students to be confident mathematicians that can explain and represent their thinking accurately, effectively and efficiently.

5. THE DISTRIBUTIVE PROPERTY

To “distribute” means to divide something or give a share or part of something.

The distributive law of multiplication over basic arithmetic, such as addition and subtraction, is known as the distributive property. It is also known as distributive law of multiplication, it's one of the most commonly used properties in mathematics.

When you distribute something, you are dividing it into parts. In math, the distributive property helps simplify difficult problems because it breaks down expressions into the sum or difference of two numbers.

According to this principle, multiplying the total of two addends by a number will give us the exact same result as multiplying each addend individually by the number and then adding them together.

Distributive Property of Multiplication over Addition: When we have to multiply a number by the sum of two numbers, we use this property of multiplication over addition.

Distributive Property of Multiplication over Subtraction: The distributive property of multiplication over subtraction is equivalent to the distributive property of multiplication over addition, except for the operations of addition and subtraction.

Fun Facts: Even though [division](#) is the [inverse](#) of multiplication, the distributive law only holds true in case of division, when the dividend is distributed or broken down into partial dividends, which are completely divisible by the [divisor](#).

The distributive property distributes complex expressions into simpler terms and thus makes problems, especially with multiple factors, easier to solve.

Distributive property can be used to [simplify](#) complex [equations](#) and problems. Experience the new way of learning math with Splash Learn, which brings an interactive platform for kids where every concept is turned into a playful session. With interesting sheets, exciting quizzes, and easy-to-understand topics, transform the way your child understands math.

Exploring distributive property in different ways

1. Prodigy

Prodigy is an adaptive, game-based learning math platform loved by more than one million teachers 150 million students around the world! It offers curriculum-aligned content from every major math topic in 1st to 8th grade, including how to:

- Use the distributive property to expand and solve expressions
- Fill in the missing numbers in equivalent expressions using the distributive property

2. Word problems

- The distributive property may not seem applicable to everyday life, but let's see it in action through some [word problems](#).

3. Arrays

Visual or hands-on manipulatives help students make sense of math and concretize abstract concepts. They're especially helpful for deepening your students' understanding of the distributive property.

By breaking down expressions into bite-sized pieces, students can tackle larger and more challenging math problems. That's where distributive property helps.

As one of the most commonly used properties, it's important to learn how to perform and apply the distributive property. Without it, clearing the parentheses wouldn't be possible. By incorporating EdTech resources, arrays, or math word problems, students should see the hands-on, practical applications of the distributive property. Did one example work more effectively to engage students and deepen their understanding? There's only one way to find out — give them a try! **Prodigy Math Game** is a game-based learning platform for math that's easy to use for educators and students alike. Aligned with curricula across the English-speaking world, your free teacher account gives you access to tools that help with differentiation, motivation and assessments.

6. TYPES OF MEDIA

There are many types of Instructional media; Visual media, Audio-Visual media, Projected media, Printed media, Computer's program, Human & the Environment, Audio media, and Physical objects.

Educational media refers to channels of communication that carry messages with an instructional purpose. They are usually utilized for the sole purpose of learning and teaching (Webcrawler, 2013).

Classification of Educational Media There are different ways to classify media. Print media, non-print media, and electronic media.

(1) Print media: These include: books, journals, magazines, newspapers, workbooks, and textbooks. These are easy to use, portable and inexpensive.

(2) Non-print media: These include: projected and non-projected media.

(3) Electronic media: These include audio media, visual media and audio-visual media, projected media and non-projected media.

(i) Audio media: These are the teaching-learning devices that appeal to the auditory sense. In other way these media can be heard alone, it carries sounds, for example audio tapes, record player, radio.

(ii) Visual media: These are the media that appeals to the sense of seeing (eyes) or the media which can be seen, example: television, computer, white board.

(ii) Audio-Visual: It refers to those instructional materials which provide learners with audio and visual experiences by appealing to the hearing and seeing senses at the same time, for example television, video tapes, and closed circuit television (CCTV).

(iv) Projected media: Projected media belong to a group of instructional resources which can only be accessed by means of projecting their content on the screen or wall using a projector machine specially designed for the purpose Gwarinda(2002). Projected media are usually a combination of software and hardware. These require light source for projection, for example, film projector slides, and so on.

(v) Non-projected media: These aids do not require any projection on screen. These do not require light source. They include 3 dimensional objects, 2 dimensional objects, prints, charts, models and so on.

Multimedia

Multimedia combines five basic types of media into the learning environment- text, video, sound, graphics and animation, thus providing a powerful new tool for education. Multimedia includes a combination of text, audio, still images, animation, video, or interactivity content forms (Vaughan, 1993). Multimedia is usually recorded and played, displayed, or accessed by information content processing devices, such as computerized and electronic devices, but can also be part of a live performance. Multimedia devices are electronic media devices used to store and experience multimedia content. Multimedia is distinguished from mixed media in fine art; by including audio, for example, it has a broader scope (Vaughan, 1993).

Relevance of Educational Media in Teaching & Learning Process

Teachers are the pillars of education system. Quality of teachers helps in maintaining high quality of teaching and learning process. This in turn affects the high outcomes. Leaders set an example for implementation of curriculum in the classroom and beyond. This implementation has a profound effect on the way technology is utilized and embedded in teaching & learning process.

There are various types of educational media and multimedia technology currently utilized in teaching and learning processes which are: computer system, microphone, mobile device, interactive white board, digital-video-on demand, online media stream, digital game, pod-cast and so on.

Educational media help from introduction of lesson to evaluation of lesson i.e. from start to end in a teaching learning process. It provides concrete experiences which serves as a basis of thinking, reasoning and problem solving. It increases the initial learning and permanency of learning.

Computer plays a key role in the modern education system. Students find it easier to refer internet than searching information from fat books. The process of learning has gone beyond from textbooks to internet with a large amount of information.

Education has no longer limited to classrooms, it has reached far and wide. Location far away from each other has come closer due to computer and internet. Presentation of information has become easier; it has facilitated audio-visual representation of information and makes the process of teaching and learning easier and interactive.

Computer system allows a teacher in the classroom to demonstrate a new lesson, animate, present new materials, illustrate how to use new programs and show new websites. In a classroom with noise, microphones can be used in teaching learning process.

Mobile devices such as clickers or smartphones can be utilized in enhancing feedback activities during and after instruction delivery by the teacher. An interactive white board provides touch control of computer applications which enhance the experience in the classroom by displaying visuals that can be viewed on a wider screen by learners'. This assist in visual learning, and

interactivity for learners to draw, write or manipulate images on the interactive whiteboard. The digital video eliminates the need for in-classroom hardware players and allows teachers and learners to access video clips immediately without the internet access. Online media streams can enhance streamed video websites for classroom teaching and learning processes. The digital game motivates the learners in learning a particular concept at hand and its use is increasing every day. Podcast (Popular medium specifically for accessing and assimilating audio and video information) is relatively new invention. Video Conferencing -Integration of computer with telecommunication systems.

7. SELECTION OF MEDIA

Media selection is an integral part of the Instructional Systems Design process. In that role, media selection ensures that a specific instructional medium can support the attainment of a given learning objective.

It gives equal attention to educational and operational issues; it will identify critical differences between different media and technologies, thus enabling an appropriate mix to be chosen for any given context; it is easily understood, pragmatic and cost-effective; it will accommodate new developments in technology.

Here are a few practical guidelines on the selection and use of instructional media:

1. Text is better than video and audio when the topic is complex. ...
2. Make text understandable and clear, avoid excessive wording.
3. Minimize reading. ...
4. Develop a good writing style, following basic writing principles

Models for Media Selection

- Students
- Ease to use
- Cost/ time
- Teaching media characteristics/ instructional strategies
- Interaction
- Organizational issues
- Networking
- Security and privacy

A model for technology selection and application is needed therefore that has the following characteristics:

- It will work in a wide variety of learning contexts;
- It allows decisions to be taken at both a strategic, institution-wide level, and at a tactical, instructional, level;

- It gives equal attention to educational and operational issues;
- It will identify critical differences between different media and technologies, thus enabling an appropriate mix to be chosen for any given context;
- It is easily understood, pragmatic and cost-effective;
- It will accommodate new developments in technology.

8. FOUR STEPS PROCESS

Teachers can differentiate instruction through four ways: 1) content, 2) process, 3) product, and 4) learning environment.

1. Content

Fundamental lesson content should cover the standards of learning set by the school district or state educational standards. Some students in a class may be completely unfamiliar with the concepts in a lesson, some students may have partial mastery, and some students may already be familiar with the content before the lesson begins. The teacher may differentiate the content by designing activities for groups of students that cover various levels.

Students who are unfamiliar with a lesson may be required to complete tasks on the lower levels: remembering and understanding. Students with some mastery may be asked to apply and analyze the content, and students who have high levels of mastery may be asked to complete tasks in the areas of evaluating and creating.

Examples of differentiating activities:

- Match vocabulary words to definitions.
- Read a passage of text and answer related questions.
- Identify an author's position and provide evidence to support this viewpoint.
- Create a PowerPoint presentation summarizing the lesson.

2. Process

Each student has a preferred learning style, and successful differentiation includes delivering the material to each style: visual, auditory and kinesthetic and through words. Not all students require the same amount of support from the teacher, and students could choose to work in pairs, small groups or individually. While some students may benefit from one-on-one interaction with a teacher or classroom aide, others may be able to progress by themselves. Teachers can enhance student learning by offering support based on individual needs.

Examples of differentiating the process:

- Provide textbooks for visual and word learners.

- Allow auditory learners to listen to audio books.
- Give kinesthetic learners the opportunity to complete an interactive assignment online.

3. Product

The product is what the student creates at the end of the lesson to demonstrate the mastery of the content. This can be in the form of tests, projects, reports or other activities. Teachers may assign students to complete activities that show mastery of an educational concept in a way the student prefers, based on learning style.

Examples of differentiating the end product:

- Read and write learners write a book report.
- Visual learners create a graphic organizer of the story.
- Auditory learners give an oral report.
- Kinesthetic learners build a diorama illustrating the story.

4. Learning environment

The conditions for optimal learning include both physical and psychological elements. A flexible classroom layout is key, incorporating various types of furniture and arrangements to support both individual and group work. Psychologically speaking, teachers should use classroom management techniques that support a safe and supportive learning environment.

Examples of differentiating the environment:

- Break some students into reading groups to discuss the assignment.
- Allow students to read individually if preferred.

Merits

- Instruction is effective for high-ability students as well as students with mild to severe disabilities.
- When students are given more options on how they can learn material, they take on more responsibility for their own learning.
- Students appear to be more engaged in learning, and there are reportedly fewer discipline problems in classrooms where teachers provide differentiated lessons.

Demerits

- Differentiated instruction requires more work during lesson planning, and many teachers struggle to find the extra time in their schedule.
- Critics argue there isn't enough research to support the benefits of differentiated instruction outweighing the added prep time.

9. THE OUR TECHNOLOGIES

Use of Instructional Technology: It is a systematic method of developing, designing, using, managing and evaluating the process of teaching and learning. The purpose of instructional technology is to provide solutions to instructional problems and improve the effectiveness and efficiency of learning.

Today, a majority of academic institutions rely on technology to enhance the learning experience. In fact, it's impossible to imagine the absence of technology in classrooms today. University students often carry their laptops to class. The increasing prevalence of technology hints at a greater cultural shift. Educators and instructors can help students learn in meaningful ways by relying on digital tools.

Instructional technology is one such byproduct of the great cultural shift. As the name suggests, it refers to the use of technology for instruction or education. It's a systematic method of developing, designing, using, managing and evaluating the process of teaching and learning. The purpose of instructional technology is to provide solutions to instructional problems and improve the effectiveness and efficiency of learning.

Here are different types of instructional technology tools that have been used in classrooms over time:

- Interactive Whiteboards Allow Learners and Instructors to Manipulate Texts and Objects On a Computer Screen. It's Especially Useful for Drawing Charts and Explaining Analytical Concepts or Frameworks.
- Audio-Visual Aids Such as Colorful Display Boards Emphasize the Course Content with Greater Impact. Video Clips and Movies Are Also a Great Way to Discuss Perspectives and Build On Ideas.
- Learning Management System (LMS) Is A Software Application That Documents, Delivers and Reports Educational Courses and Materials. There Are Several E-Learning Tools and Platforms Available Online.

Through technological means, instructors can teach learners how to search and interpret information with greater efficiency.

What Are the Benefits of Instructional Technology

Instructional technology has the potential to improve learning habits in the long run. Here are some advantages of instructional technology in education that indicate its importance in current times:

1. Strong Feedback Channels

Instructional technology helps gather structured feedback through several means such as surveys and polls. Gaining structured feedback encourages instructors to incorporate changes and improve the learning experience. Instructors can use a variety of digital tools to gauge their students' progress and offer suggestions to stay on track.

2. Interactive Collaboration

Instructional technology has made social or community learning possible. Instead of learning and working in silos, digital tools allow learners to work collaboratively beyond the classroom. Not only does it encourage collaborative learning among students but also encourages them to work closely with educators.

3. Holistic Development

The use of instructional technology provides an all-round development as digital tools often support and encourage behavioral and social skills. Also known as 21st century skills, communication, creativity and collaboration skills are highly valued in the workplace. Therefore, learners prepare for the world of work as they navigate different interactive experiences.

In a world where virtual learning and online modes of education have become the new normal, various types of instructional technology have come as a boon. If instructors want to enhance their students' learning experience, here are some examples of instructional technology that can make a significant impact:

- PowerPoint Presentations, where learners Are encouraged to create and present content
- Project-based learning, where learners gain knowledge and skills by working on one problem or project for extended periods of time
- Embedding questions in videos, where students are asked to consume video content that includes questions

10. COMPUTERS IN EDUCATION

Computers in the classroom have multiple benefits for both students and their teachers.

Computers are one of the most valuable resources in a classroom because they serve so many useful functions. With computers and the internet, students today have a wealth of information at their fingertips that can help them develop their research and communication skills while preparing them for a future career in a workforce that is increasingly reliant on computer technology.

One of the most common applications of computers in education today involves the ongoing use of educational software and programs that facilitate personalized online instruction for students. Educational software like this makes it easier to differentiate instruction so that lessons meet each student's unique learning needs. These tools also provide a wealth of useful data and resources that teachers can use to work with their students in the classroom and maximize learning. Online assessments are more efficient than traditional paper testing because it allows for more immediate feedback and data.

Computers also have an important role beyond primary and secondary education classrooms. Thanks to computers and technological advancements, higher education is now more accessible than ever. Many colleges and universities offer online classes, and some even offer degree programs that can be completed exclusively online. Online classes and online degree programs

make it easier for single parents or students with heavy workloads to continue their education from the comfort of their own home and at their own pace.

The benefits of using computers in the classroom goes beyond more efficient assessment and opportunities for online learning. Mobile devices and technologies are an inevitable part of society, but that does not mean that students naturally understand how to use those technologies appropriately. Using computers in the classroom gives teachers an opportunity to teach digital citizenship skills that demonstrate ways to use technology correctly and responsibly.

Computers also help maximize student engagement. Modern students are regularly exposed to technology outside of the classroom. Most use and enjoy smartphones and other mobile devices, which is why they are more likely to engage in the learning process if it involves something to which they are already accustomed and enjoy.

Computers have revolutionized the teaching profession in multiple ways. Teachers use computers to record grades, calculate averages, manage attendance and access data on student performance in online programs and assessments. Computers have also made it easier for teachers to vary their instructional delivery. Instead of lecturing at the front of the room for an entire class period, teachers can incorporate technology into their lessons to keep students engaged while appealing to a variety of learning styles. From using computers to create presentations on a topic to showing video clips that complement the lesson at hand, technology helps teachers make the content easier for students to understand.

11. EDUCATIONAL PRINT MATERIA

Printed materials mean any publication, documents, or record including, but not limited to, the following: newspapers, magazines, books, photographs, drawings, prerecorded magnetic audio tapes. For example: Store flyers, calendars, signs, charts, menus, phonebooks labels, alphabets posters.

Printed educational materials are widely used dissemination strategies to improve the quality of healthcare professionals' practice and patient health outcomes. Traditionally they are presented in paper formats such as monographs, publication in peer-reviewed journals and clinical guidelines

Objectives

To assess the effect of printed educational materials (PEMs) on the practice of healthcare professionals and patient health outcomes.

To explore the influence of some of the characteristics of the printed educational materials (e.g. source, content, format) on their effect on healthcare professionals' practice and patient health outcomes.

Advantages

Extremely portable: print materials can be used in any location.

High comfortable: Mostly students are very comfortable using print materials to learn.

Cost effective: Print materials can be created and duplicated with little expense.

Readily available: Many distance learning courses can take advantage of existing textbooks, thus saving the time and expense of creating custom materials.